2.5 Describe data types

Variables can be assigned a specific data type by the programmer. A variable's data type determines the type of value that can be stored in the variable and the operations that can be performed on it.

Some of the most common data types used in computer programs are:

- Integer
    - Used for whole number.
    - It can be positive, negative, or zero.
    - Tthere is no fractional or decimal part.
    - Examples: 5, 0, -12
- Floating-Point
    - Used for numerical values that can contain a decimal point or fractional part
    - Example: 2.345, -4.5, 10.0
- String
    - Used for text.
    - String values are usually surrounded by quotation marks.
    - Examples: "Hello", “John”, “10”
- Boolean
    - Can be one of two values: True or False.
    - We will learn more about Boolean values later in this course.

For example, if you want to create a variable to store a person's name, you will use a String data type. A person’s age might be stored as an Integer, while a value such as price might be stored using a Floating-Point data type.

Examples

Determine the correct data type.

a) 0.0065

b) 1234.

c) "false"

d) false

e) 1.99

f) 25

g) “25”

h) -47

i) “A102”

Write an assignment statement for each of the following.

a) Calculate the average of 3, 9, and 12.

b) The cost of attending a movie is $12 for an adult and $7 for a child. Calculate the total cost for a group of x adults and y children to attend the movie.

c) A program stores a person's name, "Alex", in a variable called name. Write the computer assignment statement.